

# Fundamentals of Linear Control:

## A Concise Approach

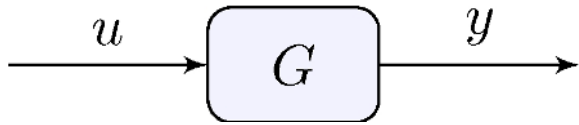
Maurício C. de Oliveira

**Chapter 1: Introduction**

[linearcontrol.info/fundamentals](http://linearcontrol.info/fundamentals)

# 1.1 Models and Experiments

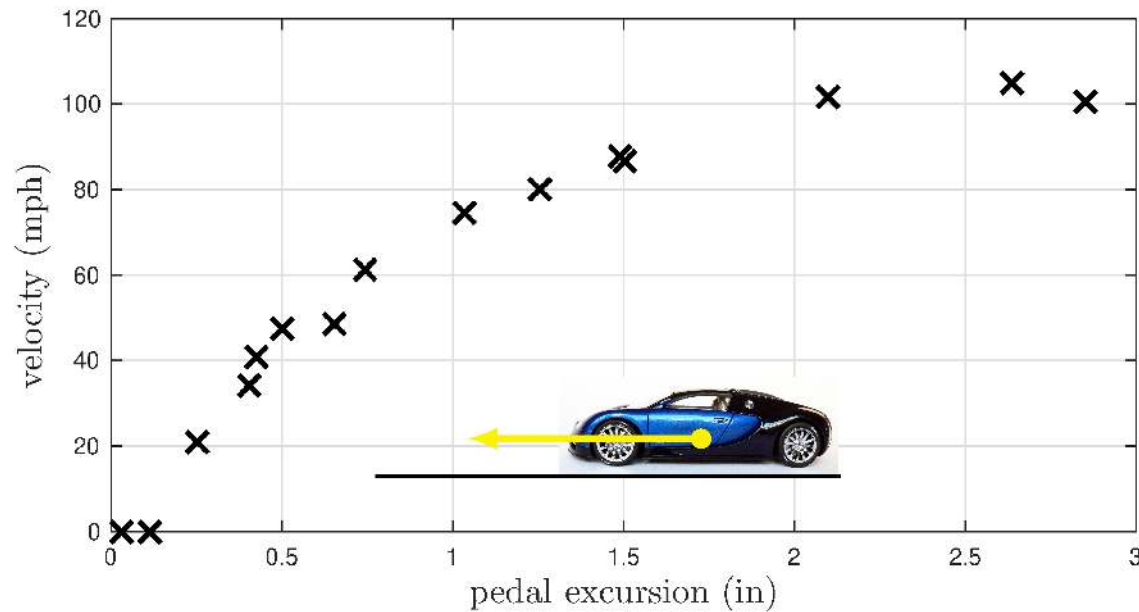
## Block diagram



- Input:  $u$
- Output:  $y$
- System:  $G$

# A simple static model of a car

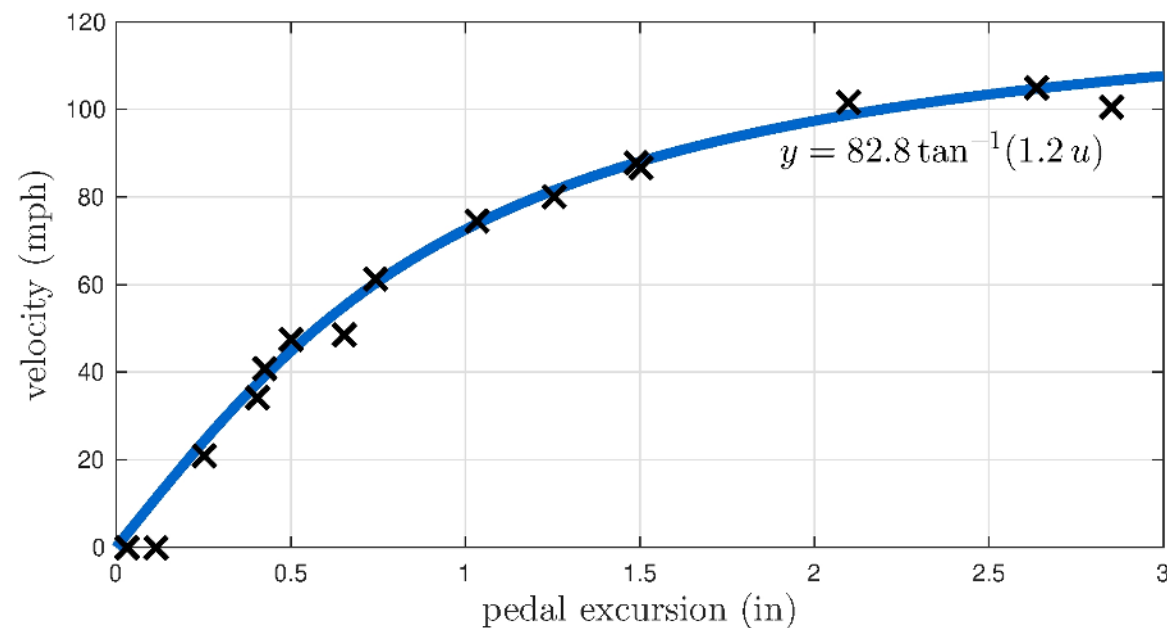
- Input: pedal excursion, in inches
- Output: terminal velocity, in mph
- System: a car *and* the experimental conditions



# A simple static model of a car

## Mathematical models

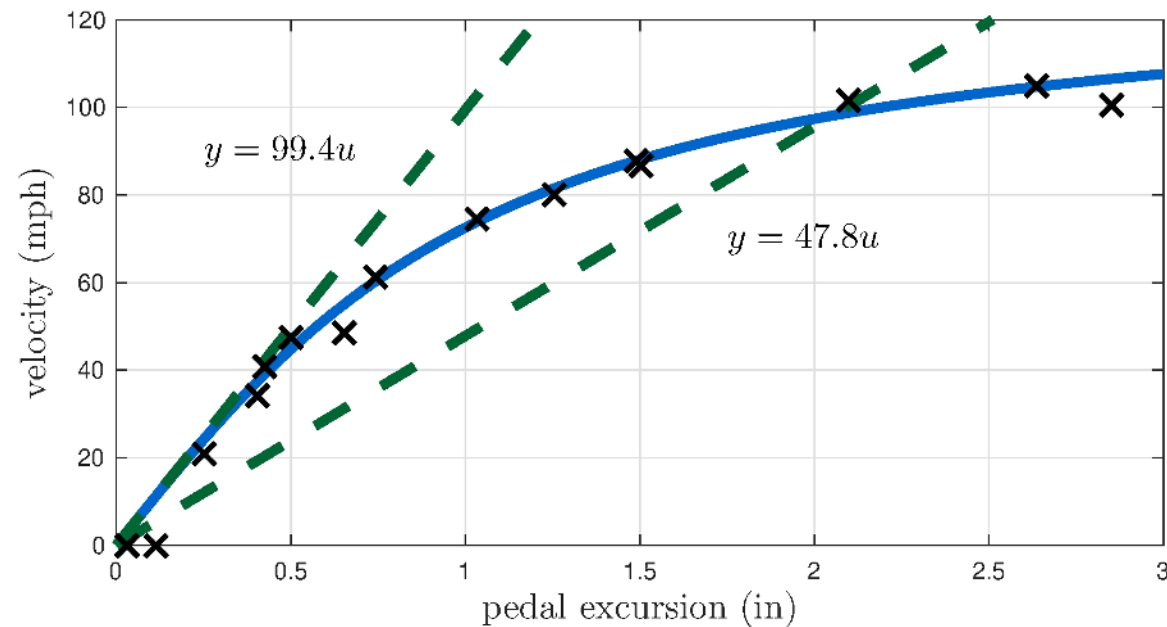
Curve fitting:  $y = \alpha \tan^{-1}(\beta u)$  (nonlinear)



# A simple static model of a car

## Mathematical models

Curve fitting:  $y = \gamma u$  (linear)



# 1.2 Cautionary Note

## A model

- should not be confused with *reality*
- does not need to be *correct*
- is not to be *believed*

## Models can be

- *static*: depends on the current value of the input
- *dynamic*: depends on the current *and* past values of the input

## In this book

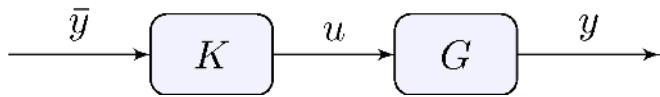
- *signals*: continuous functions of time
- *dynamic models*: ordinary differential equations

## 1.3 A Control Problem

### Tracking

Given a **target terminal velocity**,  $\bar{y}$ , design a system, **the controller**, that is **capable to command** the accelerator pedal of a car, **the input**,  $u$ , **to produce** a terminal velocity, **the output**,  $y$ , **equal to the target velocity**?

# 1.4 Solution without Feedback



System model

$$y = G(u)$$

Open-loop controller

$$u = K(\bar{y})$$

Solution

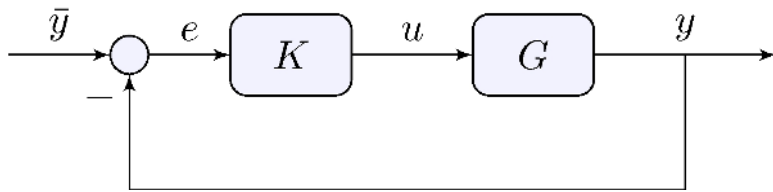
$$K = G^{-1}$$

Controller is a function of the model

Uncertainty in the model increases uncertainty in the solution



# 1.5 Solution with Feedback



System model

$$y = G(u)$$

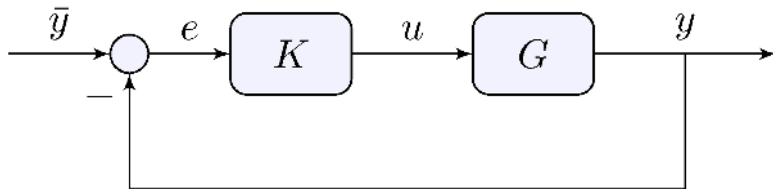
Feedback controller

$$u = K(\bar{y}, y)$$

Most common feedback controller

$$u = K(e), \quad e = \bar{y} - y$$

# Solution with Feedback



## Feedback analysis in the linear case

$$y = G u, \quad u = K e, \quad e = \bar{y} - y$$

Eliminate  $e$  and  $u$ :

$$y = G K e = G K (\bar{y} - y) \quad \implies \quad (1 + G K) y = G K \bar{y}$$

If  $G K \neq 1$

$$y = H \bar{y}, \quad H = \frac{G K}{1 + G K}$$

# Solution with Feedback

## Feedback analysis

$$y = H\bar{y}, \quad H = \frac{GK}{1 + GK}$$

$K$  large means  $H \approx 1$

Choice of  $K$  does not depend directly on value of  $G$

## Control signal

$$u = Ke = K(\bar{y} - y) = K(1 - H)\bar{y} = \frac{K}{1 + GK}\bar{y} = \frac{1}{K^{-1} + G}\bar{y}$$

$K$  large means  $u \approx G^{-1}\bar{y}$

Feedback loop *learns*  $G$ !

## Effect of $G$ , $K$ on $H$

| $G$  | $K$    |        |        |        |        |
|------|--------|--------|--------|--------|--------|
|      | 0.02   | 0.05   | 0.5    | 1      | 3      |
| 47.8 | 0.4888 | 0.7050 | 0.9598 | 0.9795 | 0.9931 |
| 73.3 | 0.5945 | 0.7856 | 0.9734 | 0.9865 | 0.9955 |
| 99.4 | 0.6653 | 0.8325 | 0.9803 | 0.9900 | 0.9967 |

$K = 0.5$  is already “large”

Value of  $H$  remain close to 1 even if  $G$  varies widely

Similar conclusions in the presence of “mild” nonlinearities

# 1.6 Sensitivity

## Transfer-function from reference to output

$$y = H(G) \bar{y}$$

open-loop:  $H(G) = GK$

closed-loop:  $H(G) = \frac{GK}{1+GK}$

## Sensitivity

$$\frac{H(\bar{G}) - H(G)}{H(\bar{G})} \approx S(\bar{G}) \frac{\bar{G} - G}{\bar{G}}, \quad S(G) = \frac{G}{H(G)} H'(G)$$

open-loop:  $S(G) = 1$

closed-loop:  $S(G) = \frac{1}{1+GK}$

Large  $K$  reduces closed-loop sensitivity

# Closed-loop transfer-functions

From reference to output

$$y = H \bar{y}, \quad H = \frac{GK}{1 + GK}$$

From reference to tracking error

$$e = \bar{y} - y = (1 - H) \bar{y} = S \bar{y}, \quad S = \frac{1}{1 + GK}$$

Complementarity:  $S + H = 1$

Large  $K$  improves tracking  $S \rightarrow 0, H \rightarrow 1$

# 1.7 Disturbances

## UNITED STATES PATENT OFFICE.

ORVILLE WRIGHT AND WILBUR WRIGHT, OF DAYTON, OHIO.

### FLYING-MACHINE.

No. 821,393.

Specification of Letters Patent.

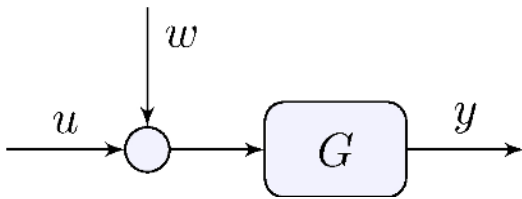
Patented May 22, 1906.

Application filed March 23, 1903. Serial No. 149,220.

55 lar manner. In either case owing to the varying conditions to be met there are numer-

ous disturbing forces which tend to shift the machine from the position which it should occupy to obtain the desired results. It is the chief object of our invention to provide 60 means for remedying this difficulty, and we will now proceed to describe the construction by means of which these results are accomplished.

# Disturbances

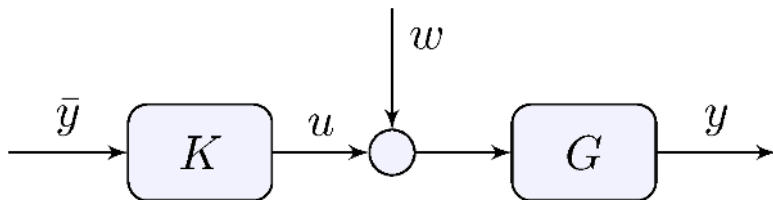


## Input disturbance

$$y = G(u + w)$$



# Disturbances



## Open-loop

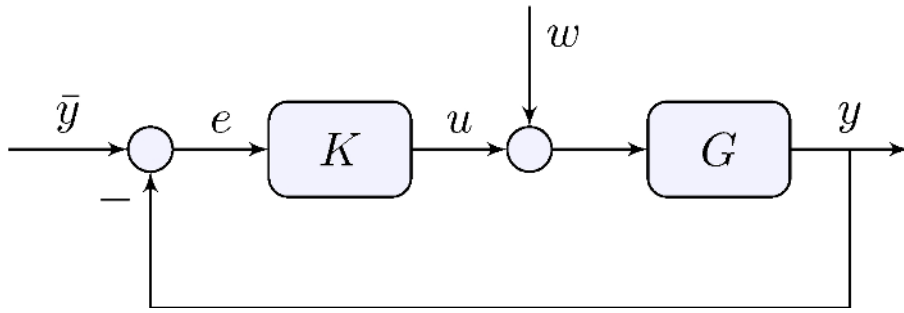
$$e = \bar{y} - y = \bar{y} - G(K\bar{y} + w) = (1 - GK)\bar{y} - Gw$$

If  $K = G^{-1}$

$$e = -Gw$$

Open-loop system cannot reject disturbance

# Disturbances



## Closed-loop

$$e = \frac{1}{1 + GK} \bar{y} - \frac{G}{1 + GK} w$$

$K$  large makes both terms small

Feedback simultaneously tracks reference *and* rejects disturbances